

Unpacking the internal assessment

General advice

Calculating the overall criterion mark

Deciding on an overall mark for a criterion can seem challenging, but it is less so when a clear methodology is used.

When marking the computational solution, read the complete documentation and watch the video first to get a general impression of the work before deciding on the marks to be awarded.

Evidence for a single criterion will usually appear in the section of the documentation dedicated to that criterion, but note that it may also be seen in other sections. Read the documentation more than once, marking the overall task against each criterion. Similarly, watch the student's video at least once. The video can give its own particular impression of the computational solution and provide evidence for awarding marks for criterion D.

A best-fit approach must be used to decide the appropriate mark to award for each criterion. The overall mark awarded for a criterion is **not** an arithmetic mean of the different strands. Rather, it is a holistic judgement reflecting the overall standard of work the student has demonstrated for that criterion.

The level descriptors of the criterion appear as bullet points within the markbands. For example, in criterion A, "The response **outlines** a problem scenario" is in the markband 1–2. Read the level descriptors for each criterion (starting with the lowest level) until you arrive at a descriptor that most appropriately describes the level reached by the student's work.

If a piece of work seems to be between two descriptors, read both descriptors again. Choose the descriptor that more appropriately describes the student's work.

Consider how well the application of command terms has been addressed, as described in the *Computer science guide*.

Once all the level descriptors have been considered as well as the markbands in which they appear, a holistic judgement can then be made about which markband the student's performance falls into for

Unpacking the internal assessment

the criterion in question. Once the markband has been identified it can then be decided which mark within the markband is most appropriate.

It is not necessary that all level descriptors within a single markband to be met for a mark within that band to be awarded.

Mark positively—give credit for what the student has done; do not penalize what they could have done or should have done. Record only whole numbers when giving marks; partial marks (fractions and decimals) are not acceptable.

Evidence of programming

During the development of the computational solution, teachers should help students to adopt good programming practices, including the use of meaningful names for variables and functions, consistent indentation and adding comments to important sections of the code (e.g. pre- and post-conditions of procedures and annotations on complex segments of code).

In criterion D (development), students should highlight techniques they have used, including excerpts of their source code together with annotations to explain their algorithmic thinking and a separate justification of its use.

Evidence of programming should include clear screenshots of key sections of the computational solution source code, highlighting specific features or algorithms.

Communication style and documentation length

The computational solution consists of three parts.

- The documentation
- The video
- The appendices

Unpacking the internal assessment

Remember that the documentation and video are used for marking the computational solution, while the appendices is where additional information that is not marked is submitted. Students need to be clear that the appendices must include the full source code for the solution.

The documentation

The documentation needs a title and contents page, and must present a section for each of the five criteria. It is recommended that section titles are simply “Section A”, “Section B”, and so on, aligning with the lettering of the criteria. Each section should deal with that part of the computational solution and the relevant criterion. It will be helpful if students understand clearly how each criterion relates to the computational thinking process.

The total word count for the documentation should not exceed **2,000 words** and must be stated on the front page. Suggested word limits for each section of the documentation are given in the *Computer science guide*. Diagrams and code with associated comments (embedded in the code or given as annotations) are not included in the word count. However, only necessary diagrams are required for the solution. Including additional diagrams that duplicate work or do not add to the solution's development should be avoided wherever possible.

Excerpts of code and associated comments should be included as evidence of the techniques used to code algorithms and the reasons for their use. These excerpts should be chosen carefully, and a reason should be given for each of them being in the documentation.

The video

It is suggested that students plan the video before recording (e.g. via a storyboard or a simple written plan), to make the most of the five minutes available to them. It is also recommended that students aim to keep the video below 500 MB.

The video must demonstrate the computational solution's functionality and give examples of the testing strategy. The audiovisual quality of the video must be sufficient to demonstrate these; however, students need to be clear that they are not being marked on this quality, but rather on what the video shows. Students should also note that the video is not intended to address the complexity or code of the product, nor possible extensions of the product.

Unpacking the internal assessment

A simple, single test-run of the product will often be sufficient for a student to demonstrate its functionality in the video, followed by some examples of the testing strategy. A straightforward screen recording can often demonstrate these, though it is also recommended that the video comes with commentary, either spoken or as on-screen text. Students are permitted to speed up the video only for specific circumstances (e.g. to show a lengthy sequence of data entry). They should not speed up the video to cram in more information overall, (i.e. by speeding up the rate at which written text appears on screen such that it cannot be read without pausing the video).

Teachers should discuss with students the most effective tools available for recording their computer screens. Different operating systems may have built-in screen recording software, or students can use free screen-recording software available online.

The appendices

The full source code of the computational solution must be included as an appendix. While this code will not be run or marked, it is an important reference for the excerpts of code included in the documentation. For example, the examiner may look at it to clarify ambiguities in the product documentation.

The appendices can also include data and other information referred to in the documentation where students feel this is not sufficiently important to include in the documentation itself—i.e. the part that will be marked. Students need to be clear that no marks are awarded for any content presented in the appendices. For example, the appendices could also include a bibliography if research has been undertaken in computer science (or another domain of knowledge) that has been referenced in the main documentation. However, the presence of this bibliography and its references would not result in higher marks being awarded in the main documentation, though the examiner might find it useful to see these full references.

